

Durability of basalt FRP reinforcing bars in alkaline solution and moist concrete environments

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HIGHLIGHTS

- BFRP bar specimens were conditioned in alkaline solution and moist concrete.
- The tensile strength and microstructural characteristics of the BFRP bars were evaluated.
- The impact of varying the conditioning environment was elucidated.
- Master curves for service life prediction of conditioned BFRP bars were developed.

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ABSTRACT

The durability performance of basalt fiber-reinforced polymer (BFRP) reinforcing bars conditioned for 3, 6, and 9 months at temperatures of 20, 40, and 60 °C was investigated. BFRP specimens were exposed to two environments: moist concrete and an alkaline solution simulating the concrete pore solution. Correlations among tensile strength, moisture uptake, matrix digestion, and microstructure characteristics were furnished. The degree of deterioration was more sensitive to the conditioning temperature than the conditioning duration. The tensile strength retentions of the specimens conditioned in the alkaline solution at a temperature not exceeding 40 °C were comparable to those conditioned in the moist concrete environment irrespective of the conditioning duration. Conditioning in the alkaline solution at 60 °C for a duration ≥ 6 months had a more detrimental effect on the microstructure and strength of the BFRP than encasing in moist concrete. After 9 months of exposure at 60 °C, the specimens conditioned in the alkaline solution lost 29% of the initial tensile strength while those encased in moist concrete lost only 15%. The intensified tensile strength reduction caused by the former conditioning scheme was due to disintegration of the matrix and degraded fiber-matrix interfacial bond caused by a higher moisture uptake and development of more hydroxyl groups during conditioning. The accelerated aging test results were utilized to develop a master curve for service life prediction of BFRP bars.

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1. Introduction

For the past two decades, studies have investigated the utilization of fiber-reinforced polymer bars as noncorrosive alternative reinforcements. Despite their numerous advantages, including corrosion resistance, high strength-to-weight ratio, and electromagnetic neutrality [1–3], their adoption by the construction industry has been hindered for various reasons. For instance, carbon FRP (CFRP) bars have shown excellent performance characteristics, but at a much higher cost than conventional steel. On the other hand, glass FRP (GFRP) bars have been associated with a

lower cost, but are susceptible to degradation in alkaline environment. One means of overcoming these limitations was the introduction of basalt FRP (BFRP) materials, which offer an enhanced chemical resistance to alkaline environment and improved fatigue performance at a much lower cost than that of CFRP [4]. The aging-related degradation in the tensile properties of FRP composite bars under different environmental conditions are highly dependent on the manufacturing procedure, chemical composition of the matrix, characteristics of the interface, and interfacial imperfections that may develop during the manufacturing process. Degradation of FRP typically occurs through hydrothermal action on the composites. Matrix softening due to hydrolysis reduces the tensile strength and stiffness of the composite material. The reaction with the coupling agents breaks down the chemical bonds at the

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fiber-matrix interface thus resulting in a separation between the fiber and matrix. Leaching of materials at the fiber-matrix interface and loss of fiber structural network could also occur due to ingress of moisture in the presence of stress and high temperatures [5]. As such, improved understanding of the long-term durability performance of FRP is required prior to its wide adoption by the construction industry.

Over the past ten years, few studies have investigated the durability performance of BFRP bars exposed to aggressive environments under accelerated aging techniques. Generally, the service life of BFRP bars in concrete has been simulated using alkaline concrete-pore solutions. Benmokrane et al. [6], Ali et al. [7], and Elgabbas et al. [8] subjected basalt FRP bars to alkaline solution for up to 5000 h at 60 °C. Research findings showed that alkaline conditioning caused significant deterioration to the physical and mechanical properties of BFRP bars made with vinylester resin. Less damage was reported for BFRP bars with an epoxy resin. In fact, the shear, flexural, and inter-laminar shear strength of such BFRP bars decreased by 12, 19, and 21%, respectively, upon conditioning for 5000 h. Corresponding microstructure evaluation tests showed that the major deterioration mechanisms were attributed to a hydrolysis reaction and were in the form of delamination and debonding at the fiber-matrix interface and an increased formation of hydroxyl groups in the resin. Another study [9] on BFRP bars immersed in an alkaline solution for 9 weeks at 55 °C noted a 40% loss in tensile strength. Moisture absorption results showed that deterioration was indeed due to degradation of the matrix and fiber-matrix interface. A comparison of these BFRP bars with GFRP equivalents showed similar performance during early conditioning, but the former maintained superior properties over extended conditioning durations. In other work, it was reported that BFRP bars conditioned in alkaline solutions with pH ranging from 9 to 13 for up to 5000 h at temperatures between 20 and 60 °C retained at least 55% of the initial tensile strength [10]. More recently, Guo et al. [11] conditioned BFRP bars in simulated seawater sea sand concrete environment for 6 months at temperatures up to 60 °C. Such conditioning induced severe deterioration to the bars, owing to a chemical reaction between aluminum, iron, and magnesium in the BFRP with the chlorides present in seawater. Microstructure analysis emphasized the deterioration to the fiber-matrix interface, formation of microcracks, and the development of more hydroxyl groups in the matrix.

The effect of alkaline solutions, simulating the concrete pore solution, on the performance of BFRP bars have been predominantly investigated. Nevertheless, some studies examined their properties upon exposure to different environmental conditions. A study by Wang et al. [12] examined the properties of BFRP bars exposed to acidic solutions. The authors noted a strength retention of 81.6% after subjecting the bars to an acidic solution (pH = 2) for 9 weeks at 55 °C. The developed prediction model estimated that the conditioned samples would require 200 years for 20% of the strength to be lost. Clearly, BFRP reinforcing bars are more resistant to an acidic environment than an alkaline one. Other work by Shi et al. [13] investigated the degradation of BFRP tendons subjected to salt solution for up to 63 days at temperatures of 25 and 40 °C, after which creep load was applied. As such, the combined effect of conditioning under sustained load was not attained. It was reported that at least 91% of the initial tensile strength was retained and no severe deterioration was induced by the sole exposure to the salt solution. In fact, the majority of the damage was caused by the subsequent sustained creep load. Micrographs of a scanning electron microscopy (SEM) and spectra of Fourier transform infrared (FTIR) analysis highlighted debonding of the fiber from the matrix, deterioration at the fiber-matrix interface, and hydrolysis reaction. Furthermore, a comprehensive study on the combined effect of sustained load and different simulated

corrosive solutions, including alkaline, acid, and salt solution and de-ionized water, on the durability performance of BFRP bars was conducted [4]. Experimental results showed much less severe deterioration in acid solution, salt solution, and de-ionized water compared to alkaline solution. It was also found that the deterioration was aggravated when the stress level reached 40% of the ultimate tensile strength and at an elevated temperature of 55 °C. Microstructure characterization through SEM highlighted the deterioration at the fiber-matrix interface rather than in the fibers. Also, a service life prediction model showed that BFRP samples could retain 50% of their initial tensile strength when exposed to alkaline solution for 4, 7, and 16 years at 20, 14, and 6 °C, respectively.

Despite BFRP reinforcing bars having a great potential to replace conventional steel reinforcement in concrete structures to overcome corrosion-related problems, evaluation of their mechanical properties and microstructure characteristics upon conditioning in moist concrete environment has seldom been investigated. Accordingly, this paper aims at filling this gap by providing an improved understanding of the durability performance of BFRP reinforcing bars encased in moist concrete. Results of BFRP bar specimens conditioned in moist concrete were compared to those conditioned in an alkaline solution to evaluate the applicability of using a simulated concrete pore solution rather than actual moist concrete. The durability performance of unconditioned/control and conditioned BFRP bars was evaluated through tensile strength retention and correlated to microstructure characterization results. Test results were utilized along with Arrhenius concept to develop master curves that can predict the tensile strength retention of BFRP bars in the city of Dubai.

2. Experimental setup

2.1. BFRP bar specimens

BFRP reinforcing bars made of continuous basalt fibers impregnated in epoxy resin were employed in this investigation. The BFRP bars were sand-coated and had a nominal diameter of 10.96 mm, as shown in Fig. 1. The average cross-sectional area was determined as 94.3 mm², in accordance with the procedure of ACI 440.3R [2]. The basalt fiber content, by mass, was calculated by matrix digestion using nitric acid, according to ASTM D3171 [14]. It was found to be 74.5%. The void content was determined as 0.24%, as per ASTM D3171 and D2734 [14,15]. The BFRP specimens used for tensile testing had a total length of 1200 mm to allow installation of end grips after conditioning. Additional BFRP specimens, 200 mm long each, were subjected to the same conditioning regimes as those of the 1200 mm long specimens. Samples of microstructure tests were taken from the middle 100 mm of these 200 mm long specimens. Unconditioned BFRP specimens were tested to obtain the as-received tensile strength and modulus following the test method specified by the ACI 440.3R-12 [2]. The

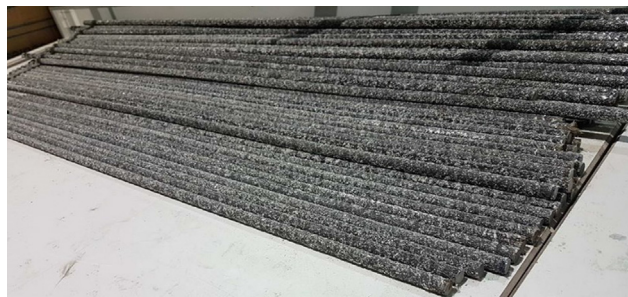


Fig. 1. BFRP bars investigated in this study.

respective tensile strength and modulus were reported as 1268 MPa and 48AGPa.

2.2. Surrounding media

The specimens were conditioned either in an alkaline solution or in a moist concrete environment. For the former, a NaOH-based alkaline solution, with a pH value of approximately 12, was utilized to simulate the pore solution of real-life concrete. The middle third of the 1200 mm long specimens conditioned in a moist concrete was surrounded by 50×50 mm concrete to simulate BFRP reinforcing bars in field conditions, as illustrated in Fig. 2. For each testing condition, three replicate specimens were utilized. The concrete used in this work was supplied by a ready-mix company with a nominal cylinder compressive strength of 40 MPa. The obtained concrete was tested for compressive and splitting tensile strengths, ultrasonic pulse velocity (UPV), and pH as per ASTM C39 [16], C496 [17], C597 [18], and E70 [19], respectively. Test results, summarized in Table 1, show that the average compressive and splitting tensile strengths were 47 and 3.2 MPa, respectively, while the pH value was approximately 12.5. The average measured UPV value was 4.7 km/s, which indicated an excellent concrete quality.

2.3. Conditioning tanks

Specially manufactured conditioning tanks with built-in heaters and thermostat were employed to maintain constant water and alkaline solution temperatures of 20, 40, and 60 °C during conditioning. The concrete-encased BFRP bar specimens were immersed in tap water whereas those of the alkaline conditioning regime were immersed in the alkaline solution (Fig. 3). The conditioning durations were 3, 6, and 9 months. The temperature of the water and alkaline solution inside the conditioning tanks was monitored on a regular basis during conditioning and any deviation in the temperature was adjusted accordingly. Results of conditioned specimens were compared to those of the unconditioned control specimen.

3. Experimental testing

3.1. Tensile strength

The BFRP specimens were retrieved carefully from the concrete prisms after conditioning. End grips were installed in preparation



Fig. 2. Concrete-encased BFRP bar specimens.

Table 1

Physical and mechanical characteristics of surrounding concrete.

Property	Unit	Standard	Value
Compressive strength	MPa	ASTM C39 [16]	47.0
Tensile splitting strength	MPa	ASTM C496 [17]	3.2
UPV	km/s	ASTM C597 [18]	4.7
pH	–	ASTM E70 [19]	12.5



Fig. 3. Test specimens in alkaline conditioning tank.

for tensile strength testing. The end grips entailed a 400-mm long steel pipe at each end with respective inner and outer diameters of 32 and 40 mm, as shown in Fig. 4(a). The inner surface of the steel pipe was roughened and thoroughly cleaned with a corrosion removal compound prior to installation. A nano-grout pourable product was used to bond to steel pipes of the end grips to the BFRP specimens. The specimens were tested to failure using a universal tensile testing machine as shown in Fig. 4(b). A uniaxial tension load was applied to test specimens in a displacement control mode at a rate of 1.5 mm/min. A data acquisition system was employed to record the applied load. The tensile strength was calculated as the ratio of the maximum applied tensile load to the cross-sectional area of the BFRP bar, measured as per ACI 440.3R-12 [22].

3.2. Moisture uptake

The moisture uptake of BFRP specimens was determined following the testing procedure of ASTM D570 [20]. Triplicate samples were used to obtain an average. The conditioned specimens were dried in an oven at 100 °C for 24 h to obtain the oven-dried mass, and the corresponding experimental moisture uptake, by mass, was determined using Eq. (1), where the initial mass and conditioned mass represent the mass of the specimen before and after conditioning, respectively:

$$\text{Moisture uptake (\%, by mass)} = 100 \times (\text{Conditioned mass} - \text{Oven-dried mass}) / (\text{Initial mass}) \quad (1)$$

3.3. Scanning electron microscopy analysis

SEM was performed using a JEOL-JSM 6390A microscope on selected BFRP specimens to assess the impact of conditioning medium, duration, and temperature on the morphology. Collected samples were polished, and sputter coated with a thin (2–5 nm)

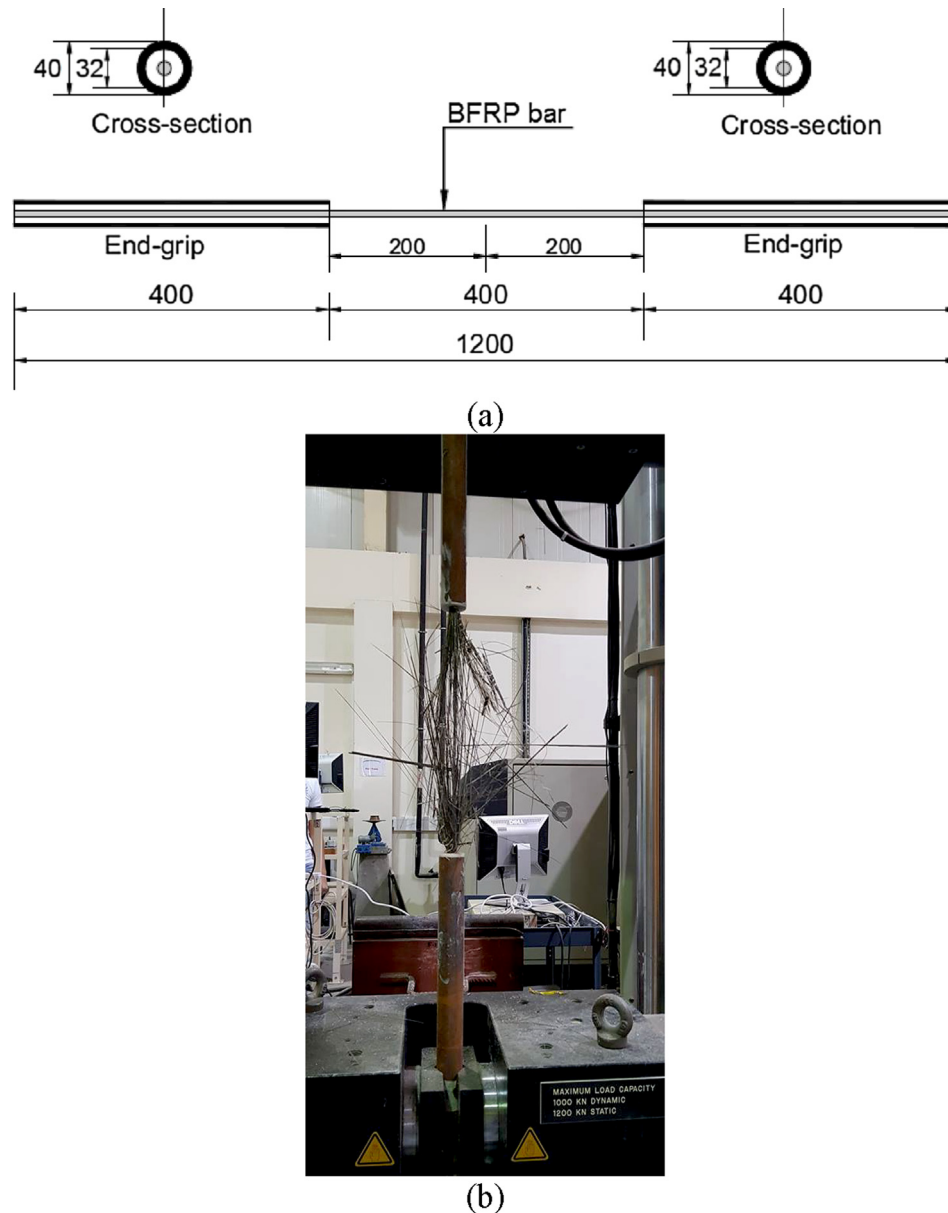


Fig. 4. Test set-up and instrumentation: (a) End grips, (b) Tensile test set-up.

99%-pure gold layer to ensure conductivity during SEM testing without masking the surface morphology. Based on the work of El-Hassan and El Maaddawy [21], it was anticipated that degradation would be identified through the micrographs as damage to the fibers, polymer matrix, and/or fiber-matrix interface. Multiple micrographs were taken for the collected samples. SEM micrographs that provided a proper representation of the microstructure and morphology of the majority of the collected samples for each testing condition were considered in the analysis.

3.4. Matrix digestion analysis

Matrix digestion with nitric acid following ASTM D3171 [14] was used to determine the fiber and matrix contents of control and conditioned BFRP reinforcing bars. Triplicate samples were used to obtain an average. The nitric acid typically dissolves and digests the epoxy resin within the BFRP samples without affecting the basalt fibers. For this test, BFRP specimens weighing an *Original*

mass of 1–2 g were prepared and combined with 50 mL of 70% aqueous nitric acid. The prepared mixture was placed in a water bath at a temperature of 80 °C for 6 h. The residual material was filtered, washed with distilled water, and oven-dried for 1 h at 100 °C, to obtain the *Fiber mass*. Eq. (2) provided the matrix mass content with respect to the total mass of the sample.

$$\text{Matrix content (\%, by mass)} = 100 \times \frac{(\text{Original mass} - \text{Fiber mass})}{(\text{Original mass})} \quad (2)$$

3.5. Fourier transform infrared analysis

FTIR spectroscopy was performed to assess the extent of degradation due to hydrolysis reaction, i.e. an increase in hydroxyl groups. Collected BFRP samples were ground into a powder with four parts of potassium bromide (KBr), by mass. A Varian 3100 FT-IR spectrometer was used to obtain a spectrum ranging from 400 to 4000 cm^{-1} at a resolution of 1 cm^{-1} .

3.6. Differential scanning calorimetry analysis

Differential scanning calorimetry (DSC) was conducted utilizing a DSC Q2000 calorimeter to determine the glass transition temperature (T_g) of control and conditioned BFRP reinforcing bars, as per ASTM E1356 [22]. Powdered BFRP samples, approx. 10 mg each, were collected then sealed in aluminum pans until testing. The DSC test procedure included a heating stage from 20 to 225 °C at a heating rate of 5 °C/min, followed by a cooling stage to 20 °C, and then a similar heating stage. The obtained DSC curves, representing two scans, were used to determine the glass transition temperatures (T_g) and consequently, assess the type of degradation mechanism. For the first scan, a decrease in the T_g of the conditioned sample compared to that of the control would indicate matrix plasticization due to moisture uptake. Yet, this plasticization would be reversible if the second scan T_g of the conditioned BFRP sample were to be in the same range as that of the control sample. On the other hand, if the second scan T_g were lower than that of the control, this would indicate an irreversible chemical degradation.

4. Experimental results and discussion

4.1. Tensile strength

The BFRP specimens failed by violent tensile rupture of fibers, as presented in Fig. 5. Tensile strength results are summarized in Table 2. The specimens are designated as six characters. The first character indicates the curing regime, where A and C refer to the alkaline solution and concrete, respectively. The second two characters signify the conditioning duration (3 M, 6 M, and 9 M), while the last three characters represent the conditioning temperature (20 C, 40 C, and 60 C). For instance, specimen A9M40C is a BFRP reinforcing bar conditioned in the alkaline solution for 9 months at 40 °C.

The tensile strength retentions of conditioned BFRP reinforcing bars are shown in Fig. 6. The specimens conditioned in the alkaline solution for 3 months at 20 and 40 °C experienced negligible reductions in the tensile strength of approximately 3.5%, whereas those conditioned at 60 °C retained 92% of the initial tensile strength. Increasing the duration from 3 to 6 months at 20 and



Fig. 5. Photos of tested BFRP bars.

40 °C for 6 months did not result in additional reductions in the tensile strength, while the tensile strength retention of the specimens conditioned at 60 °C decreased to 83%. This demonstrated that the deterioration, possibly attributed to the hydrolysis reaction, of BFRP bars intensified at the higher temperature. As a result, the chemical bonds at the fiber-matrix interface may have disintegrated, causing separation between the fiber and matrix. A further increase in conditioning duration to 9 months led to 95, 87, and 71% retentions in the tensile strength at 20, 40, and 60 °C, respectively. Increasing the conditioning temperature had more pronounced effect on the tensile strength retention than that caused by increasing the conditioning duration. This is evident through a comparison between A6M60C and A9M20C, which shows that the former retained 83% of the initial tensile strength, while the latter retained 95%. It should be noted that a significant reduction in the tensile strength is anticipated when the conditioning temperature approaches the glass transition temperature, as reported by De'Nève and Shanahan [23].

Fig. 6(b) shows the tensile strength retention of BFRP specimens encased in moist concrete. The BFRP specimens exhibited tensile strength retentions of 95, 94, and 88% after 3 months of conditioning at 20, 40, and 60 °C, respectively. Increasing the conditioning duration to 6 months had an insignificant effect on the tensile strength where 94, 90, and 88% respective tensile strength retentions were recorded. For the 9-month conditioning, concrete-encased specimens retained 91, 90, and 85% of their tensile strength. A comparison between the results reported for the alkaline-conditioned and concrete-encased BFRP specimens shows that similar tensile strengths were retained within the first 6 months. However, the impact of conditioning in the alkaline solution rather than moist concrete became more apparent at durations exceeding 6 months, with 5–14% more strength being retained when BFRP specimens were encased in moist concrete. The insignificant variation between the average tensile strength retentions of GFRP bars conditioned in wet concrete (83.6%) and those of similar bars directly immersed in alkaline solutions (84.3%) at temperatures in the range of 20 to 53 °C was reported in the literature by D'Antino et al. [24]. Nevertheless, it was recommended to avoid direct bar exposure to alkaline solutions in accelerated aging tests since in field conditions the alkaline solution would remain in the concrete pores thus covering a small area of the concrete-bar contact surface. Scheffler et al. [25] also observed that conditioning in an alkaline solution based on NaOH led to the formation of a brittle circumferential layer surrounding the fiber that reduced the effective diameter of the GFRP bars. This resulted in tensile strength reductions of 11 and 14% after 8 days of conditioning at temperatures of 20–60 °C, respectively. In contrast, no visible damage was observed when the bars were conditioned in a cement solution for up to 8 days at temperatures of 20, 40 and 60 °C. Formation of holes on the surface occurred after 8 days of conditioning 80 °C, which resulted in approximately a 50% reduction in the tensile strength.

The tensile strength retention is illustrated as a function of the conditioning temperature in Fig. 7. It is clear that BFRP specimens conditioned at the higher temperatures experienced inferior durability performance with lower tensile strength retentions. The impact of increasing the temperature on the tensile strength retention of the specimens conditioned in the alkaline solution was more pronounced at the longer duration [Fig. 7(a)]. In contrast, for the specimens conditioned in moist concrete, the reduction in the tensile strength caused by an increase in the temperature was not affected by the conditioning duration [Fig. 7(b)]. Increasing the temperature from 20 to 60 °C resulted in approximately 6.5% additional strength reduction for the specimens conditioned in moist concrete irrespective of the conditioning duration.

Table 2
Tensile strength of conditioned BFRP bars.

Sample Designation	Conditioning			Tensile strength (MPa)
	Medium	Duration (months)	Temperature (°C)	
Control	–	–	Ambient	1268.4 ± 27.9
A3M20C	Alkaline	3	20	1213.6 ± 55.9
A3M40C	Alkaline	3	40	1235.8 ± 46.4
A3M60C	Alkaline	3	60	1160.8 ± 22.8
A6M20C	Alkaline	6	20	1246.2 ± 20.5
A6M40C	Alkaline	6	40	1204.8 ± 62.6
A6M60C	Alkaline	6	60	1052.7 ± 20.0
A9M20C	Alkaline	9	20	1215.5 ± 19.4
A9M40C	Alkaline	9	40	1105.7 ± 16.1
A9M60C	Alkaline	9	60	906.7 ± 25.5
C3M20C	Concrete	3	20	1197.0 ± 17.8
C3M40C	Concrete	3	40	1186.6 ± 18.8
C3M60C	Concrete	3	60	1116.4 ± 46.4
C6M20C	Concrete	6	20	1189.9 ± 28.2
C6M40C	Concrete	6	40	1143.9 ± 10.1
C6M60C	Concrete	6	60	1121.6 ± 32.0
C9M20C	Concrete	9	20	1150.7 ± 18.7
C9M40C	Concrete	9	40	1146.6 ± 16.2
C9M60C	Concrete	9	60	1079.2 ± 15.0

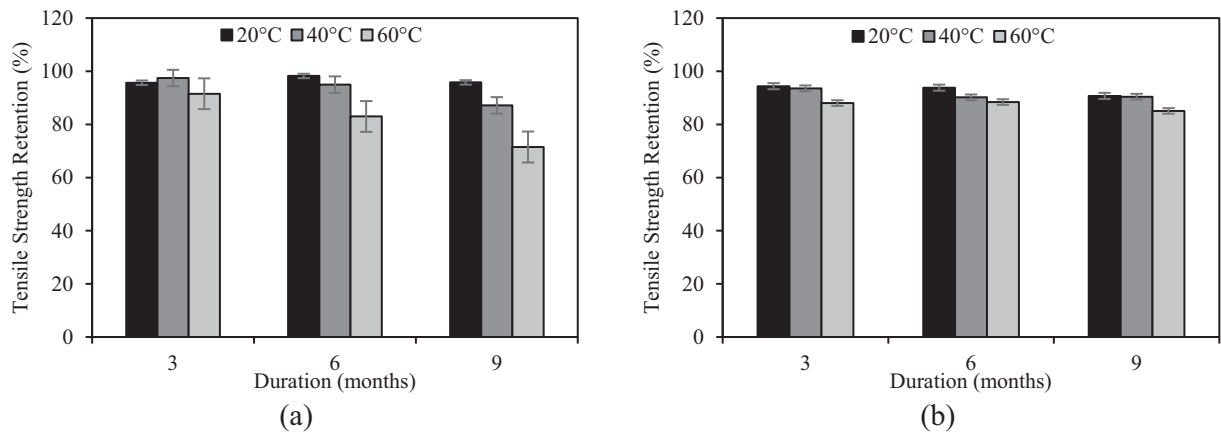


Fig. 6. Tensile strength retention of conditioned BFRP bars: (a) Alkaline solution, (b) Concrete encased.

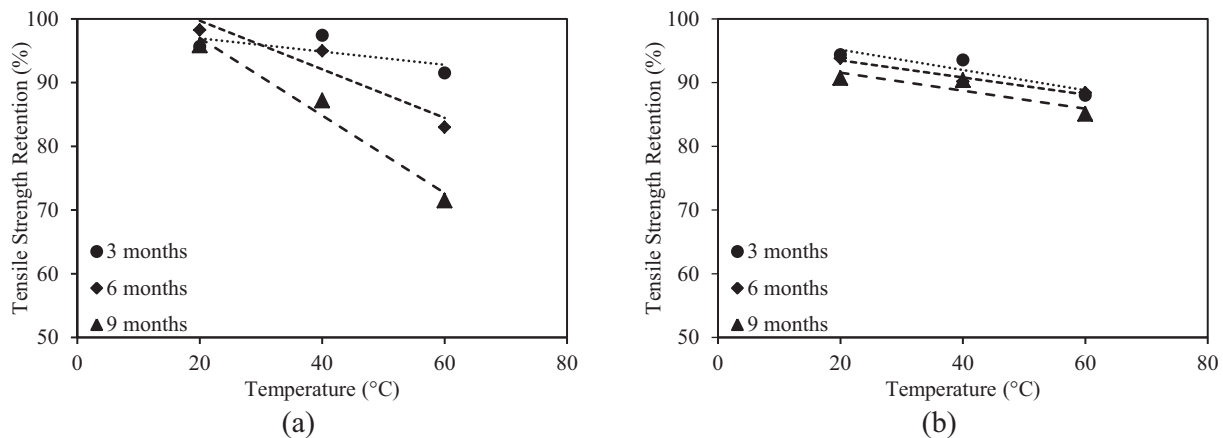


Fig. 7. Tensile strength retention versus the conditioning temperature: (a) Alkaline, (b) Concrete.

4.2. Moisture uptake

Fig. 8(a) presents the effect of conditioning duration and regime on the moisture uptake of the BFRP specimens at a constant temperature of 40 °C. Longer conditioning durations led to a higher

moisture uptake. The moisture uptake of BFRP specimens conditioned in the alkaline solution increased from 0.22 to 0.98% as the conditioning progressed from 3 to 9 months, respectively. In comparison, concrete-encased specimens showed a similar increase from 0.12 to 0.78% within the same time period (i.e. the

moisture uptake of the BFRP specimens conditioned in the alkaline solution was higher than those conditioned in moist concrete). It is worth noting that while moisture diffusion in materials typically levels out after a certain duration, the moisture uptake of BFRP specimens in the current study did not plateau, possibly because of formation of microcracks within the matrix that further promoted moisture uptake.

Fig. 8(b) displays the moisture uptake of BFRP specimens as a function of conditioning temperature. The conditioning duration, in this case, was taken as 9 months. It can be seen that higher conditioning temperatures resulted in more moisture uptake, owing to an increased diffusion rate and formation of microcracks. Indeed, BFRP specimens conditioned in the alkaline solution and those embedded in moist concrete experienced increases in moisture uptake from 0.95 to 1.04% and 0.68 to 0.86%, respectively, as the temperature increased from 20 to 60 °C. Such results provide a justification for the lower tensile strength retention recorded for the specimens conditioned at the higher temperatures, particularly those conditioned in alkaline solution at 60 °C.

4.3. SEM analysis

Cross-sectional and longitudinal micrographs of unconditioned control BFRP specimens are shown in Fig. 9(a) and (b), respectively. Proper adhesion of the matrix to the fiber and good bonding at the fiber-matrix interface are clearly noted.

After 3-month of conditioning in the alkaline solution at 40 °C, the BFRP bar seemed still intact and undamaged, as shown in Fig. 10(a). However, a separation at the interface was noticed after

6 months [Fig. 10(b)] and gap of approximately 2 μm width was noted at 9 months [Fig. 10(c)]. Such deterioration at the fiber-matrix interfacial zone could be a result of crack propagation from the matrix into the interface rather than directly into the fiber [26]. Similarly, longitudinal micrographs of the same samples showed microgaps forming at conditioning durations of at least 6 months [Fig. 11(a, b)]. Yet, at 9 months, these microgaps expanded, causing complete separation and interfacial debonding at the fiber-matrix interface, as seen in Fig. 11(c). It was also noted that the fiber surface became increasingly smoother, indicating matrix disintegration. This was primarily owed to deterioration by hydrolysis reaction, which increased with longer conditioning durations, resulting in non-uniform distribution of load among fibers [27,28] and ultimately, higher degradation of tensile properties.

The effect of conditioning temperature on the morphology of alkaline-conditioned BFRP specimens is presented in Fig. 12. Cross-sectional micrographs of samples conditioned at 20 °C present no observable deterioration [Fig. 12(a)], which confirms the limited tensile strength reduction. However, increasing the conditioning temperature to 40 °C created a small microgap along the fiber-matrix interface [Fig. 12(b)]. Yet, further increase in temperature to 60 °C resulted in a larger gap opening and more debonding at the fiber-matrix interface, as shown in Fig. 12(c). Nevertheless, the fiber remained unaffected throughout the various conditioning schemes. Longitudinal micrographs of these BFRP specimens are displayed in Fig. 13. It is clear that conditioning at 20 °C [Fig. 13(a)] induced limited damage to the BFRP, while 40 °C initiated the formation of a gap [Fig. 13(b)]. Higher conditioning temperature of 60 °C caused substantial disintegration to the matrix at

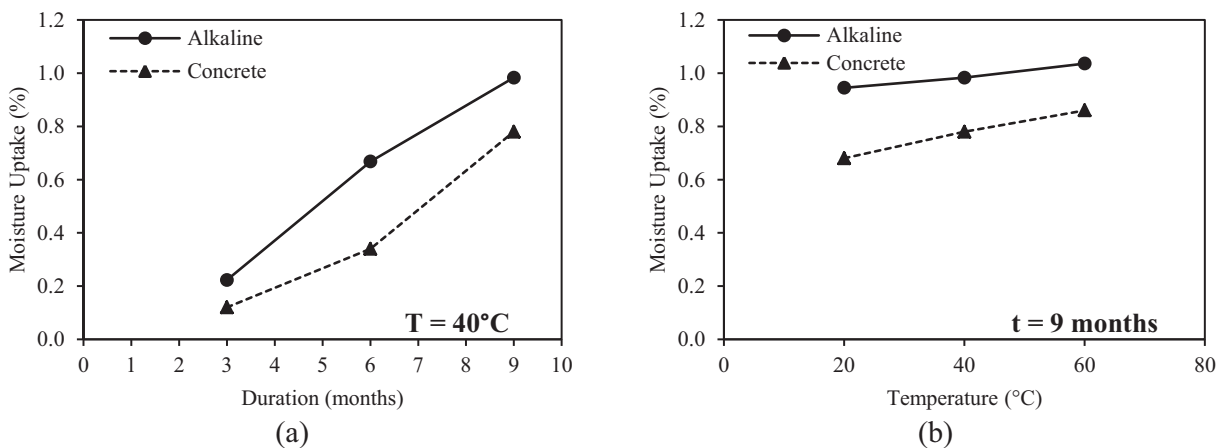


Fig. 8. Moisture uptake of conditioned BFRP bars: (a) Different durations, (b) Different temperatures.

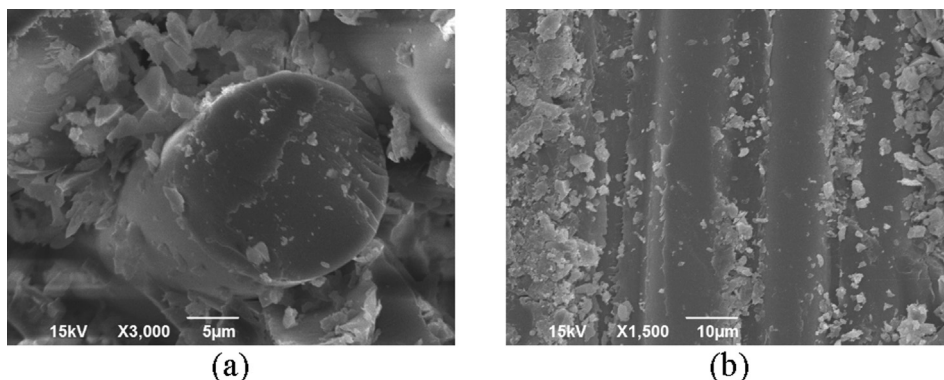


Fig. 9. Micrograph of unconditioned BFRP bars: (a) Cross-sectional, (b) Longitudinal.

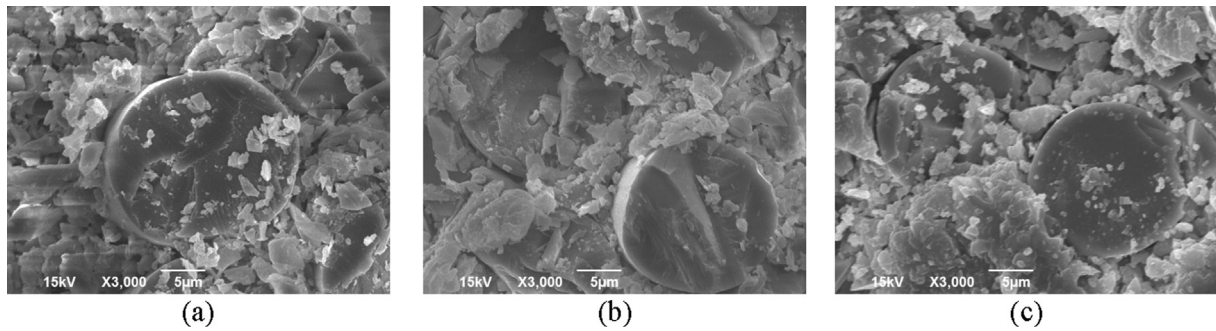


Fig. 10. Cross-sectional micrograph of alkaline conditioned BFRP bars at 40 °C: (a) 3 months, (b) 6 months, (c) 9 months.

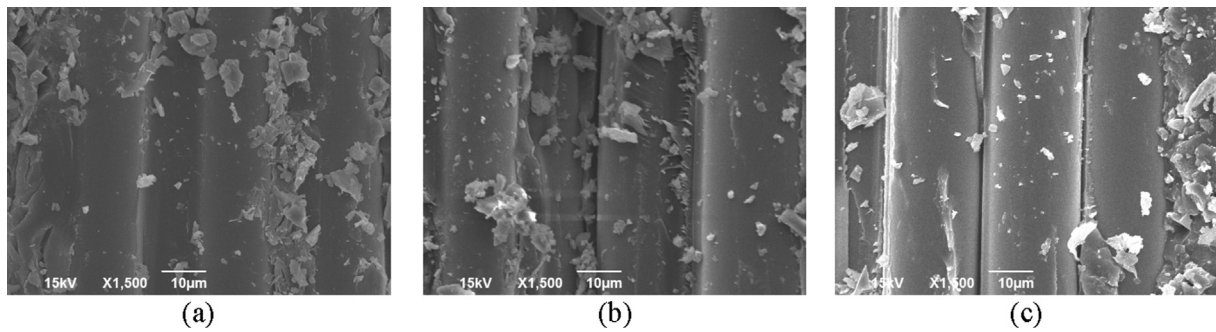


Fig. 11. Longitudinal micrograph of alkaline conditioned BFRP bars at 40 °C: (a) 3 months, (b) 6 months, (c) 9 months.

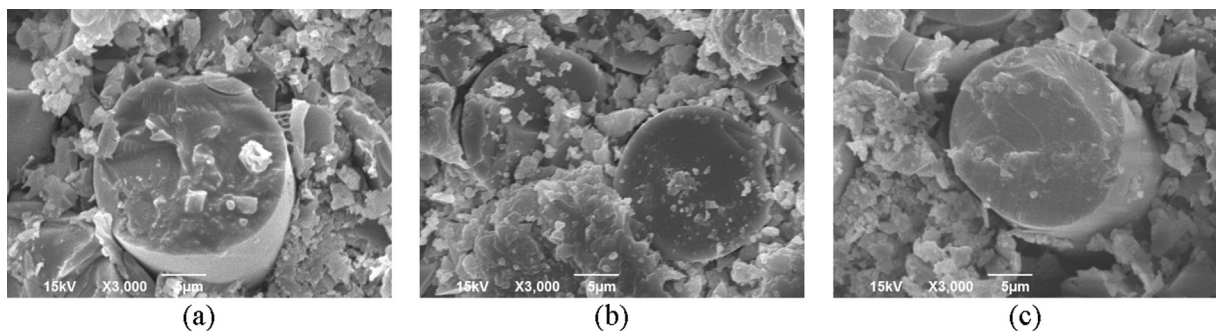


Fig. 12. Cross-sectional micrograph of alkaline conditioned BFRP bars for 9 months: (a) 20 °C, (b) 40 °C, (c) 60 °C.

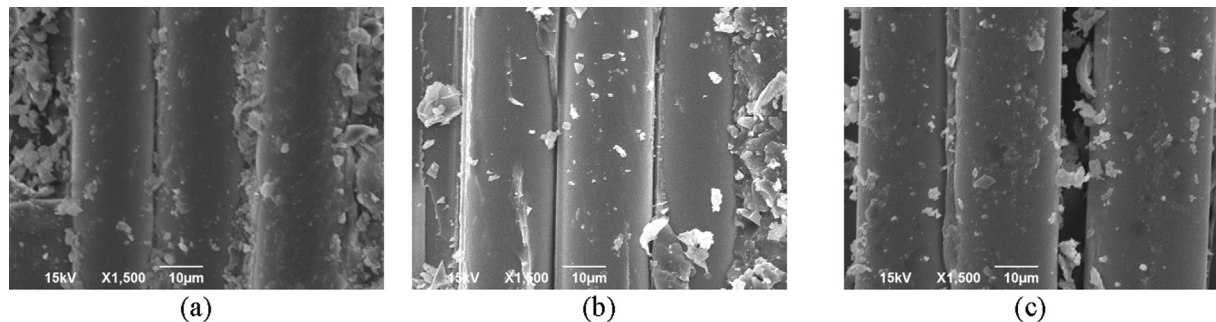


Fig. 13. Longitudinal micrograph of alkaline conditioned BFRP bars for 9 months: (a) 20 °C, (b) 40 °C, (c) 60 °C.

the fiber-matrix interfacial zone, leading to separation of the fiber from the matrix and widening of the gap to 5 μm , as presented in Fig. 13(c). Apparently, debonded and fractured surfaces may have connected to form a continuous crack. Such intense deterioration accelerated the diffusion of moisture into the microstructure by

providing more pathways and void space, which, in turn, explains the high moisture uptake and low tensile strength retention of these conditioned BFRP specimens.

Cross-sectional and longitudinal micrographs of BFRP reinforcing bars conditioned at 40 °C while encased in moist concrete are

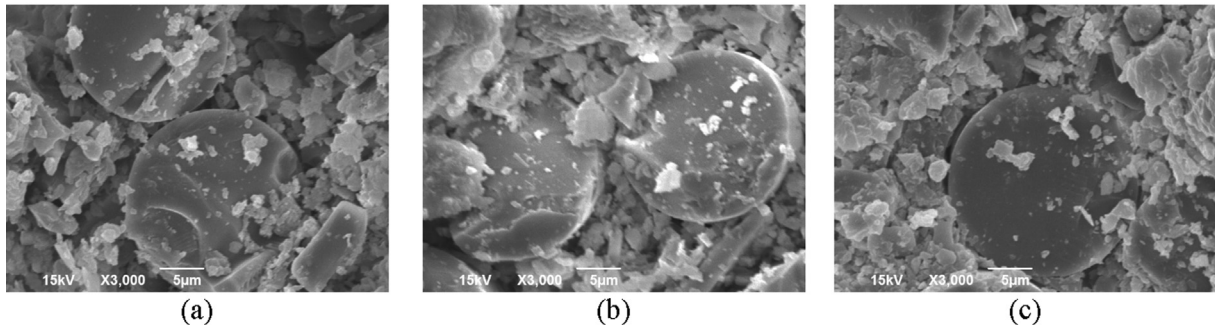


Fig. 14. Cross-sectional micrograph of concrete encased BFRP bars at 40 °C: (a) 3 months, (b) 6 months, (c) 9 months.

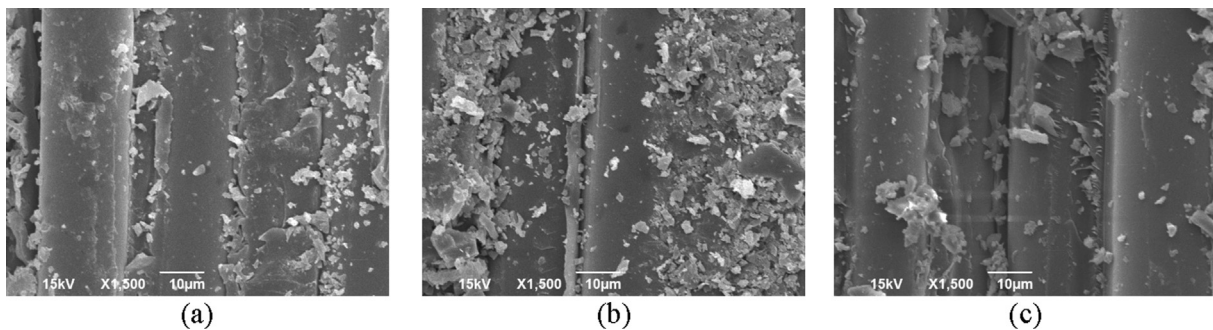


Fig. 15. Longitudinal micrograph of concrete encased BFRP bars at 40 °C: (a) 3 months, (b) 6 months, (c) 9 months.

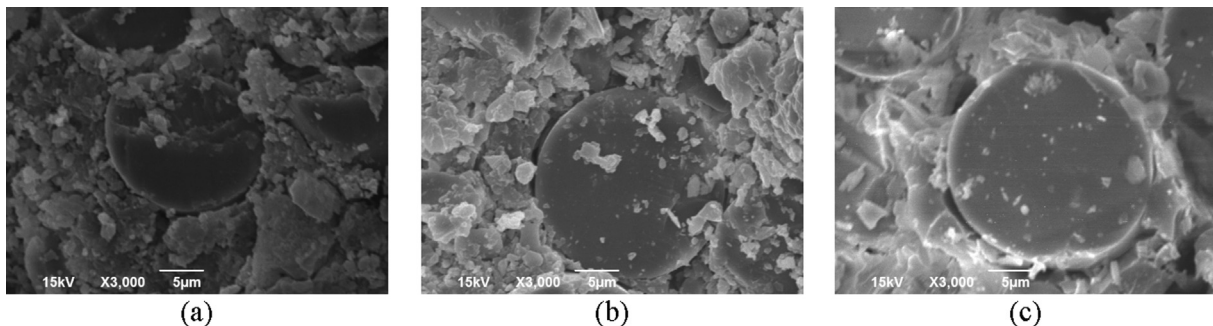


Fig. 16. Cross-sectional micrograph of concrete encased BFRP bars for 9 months: (a) 20 °C, (b) 40 °C, (c) 60 °C.

shown in Figs. 14 and 15, respectively. While a fine microgap formed after 6 months, as shown in Figs. 14(b) and 15(b), unlike alkaline-conditioned specimens, it did not widen over the next 3 months of conditioning [Figs. 14(c) and 15(c)]. This shows that the extent of damage after 6 months was too severe that further conditioning did not exacerbate the deterioration of the microstructure. This is a reflection of the tensile strength, where BFRP specimens conditioned for 3, 6 and 9 months at 40 °C retained 94, 90, and 90% of their initial strength.

Fig. 16 presents cross-sectional micrographs BFRP reinforcing bars encased in moist concrete for 9 months at different temperatures. It can be seen that a fine microcrack was incurred upon increasing the temperature from 20 to 40 °C, as shown in Fig. 16 (a, b). On the other hand, specimens conditioned at 60 °C exhibited a circumferential debonding of the fiber from the matrix, owing to an accelerated hydrolysis reaction at higher conditioning temperature [Fig. 16(c)]. Longitudinal micrographs of the same specimens are shown in Fig. 17. Results were synonymous to those of

cross-sectional micrographs, i.e. a microgap formed after 9 months at 40 °C. However, BFRP bars conditioned for 9 months at 60 °C showed a relatively smooth fiber surface [Fig. 17(c)]. These microstructure observations are in agreement with the tensile strength results, in which only 10% of initial tensile strength was lost at 20 and 40 °C after 9 months, while up to 15% loss was reported at 60 °C.

The SEM micrographs revealed that the deterioration to the fiber-matrix interface tended to be more severe when the BFRP specimens were conditioned in the alkaline solution rather in moist concrete. Microgaps typically formed in the specimens conditioned in both regimes after 6 months and at 40 °C. However, the microgaps expanded upon exposure to alkaline solution for longer durations and at the higher temperature of 40 °C. This was not the case in concrete-encased BFRP samples. Apparently, conditioning of BFRP bars in an alkaline solution for up to 6 months and at temperatures not exceeding 40 °C would have the same effect as that of the moist concrete environment at similar conditions.

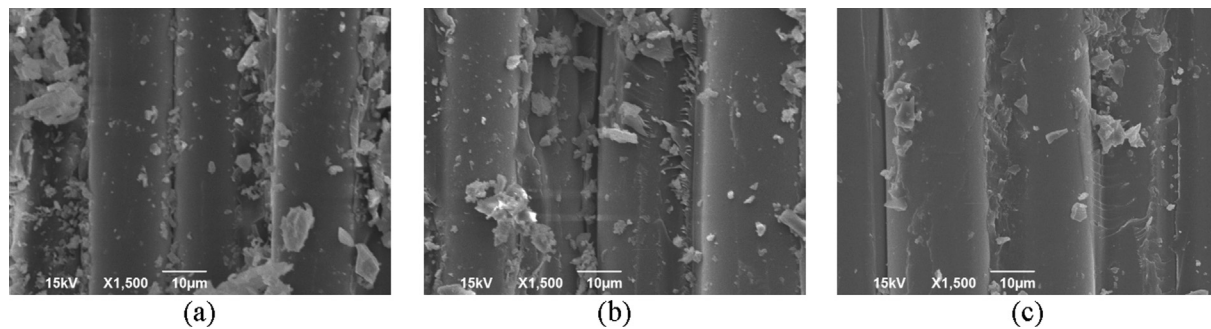


Fig. 17. Longitudinal micrograph of concrete encased BFRP bars for 9 months: (a) 20 °C, (b) 40 °C, (c) 60 °C.

Conditioning in an alkaline solution for longer durations at higher temperatures would result in aggravated/intensified deterioration relative to that caused by conditioning in moist concrete.

4.4. Matrix digestion analysis

Matrix digestion by nitric acid was used to assess the conditioning-induced deterioration and mass dissolution of BFRP reinforcing bars. A procedure was adopted in accordance with ASTM D3171 [14]. The accuracy of this methodology has been evaluated in previous work [29] by comparing experimental results obtained from this test to counterparts from the TGA test of ASTM E1868 [30]. The analysis presented herein mainly focuses on samples that showed significant loss in tensile strength and notable damage to the fiber-matrix interface using SEM. Accordingly, three samples were compared: an unconditioned control, conditioned in alkaline solution for 9 months at 60 °C, and encased in moist concrete for 9 months at 60 °C.

Table 3 shows the matrix retention of these specimens. The fiber and matrix mass contents of unconditioned control were 74.5 and 25.5%, respectively. Unlike the fiber, which was

unaffected by conditioning, as evidenced by the SEM micrographs, the matrix was more prone to deterioration. This was obvious by the reduction in matrix content to 17.9 and 18.8% upon conditioning in alkaline solution and encasing in moist concrete, respectively, representing approximately 70 and 74% of the original matrix content. Similar findings have been reported in other work focusing on conditioned GFRP bars [29,31]. A comparison between BFRP specimens exposed to alkaline conditioning and those embedded in moist concrete shows that the former conditioning scheme is more detrimental to the matrix. The degraded matrix may have dissolved in conditioning solutions or remained attached to the concrete circumferential layout surrounding the bars. Results of the matrix digestion are consistent with those of the SEM micrographs, which showed a relatively smoother fiber surface after 9-month of conditioning at 60 °C.

4.5. FTIR analysis

Figs. 18 and 19 show the FTIR spectra of BFRP bars conditioned in alkaline solution and encased in moist concrete, respectively. To evaluate the degree of hydrolysis reaction, the analysis focused on

Table 3
Matrix retention of BFRP bars after conditioning.

Sample Designation	Conditioning			Content (%)		Matrix Retention (%)	Matrix Loss (%)
	Medium	Duration (months)	Temperature (°C)	Fiber	Matrix		
Control	–	–	Ambient	74.5	25.5	–	–
A9M60C	Alkaline	9	60	82.1	17.9	70.2	29.8
C9M60C	Concrete	9	60	81.2	18.8	73.7	26.3

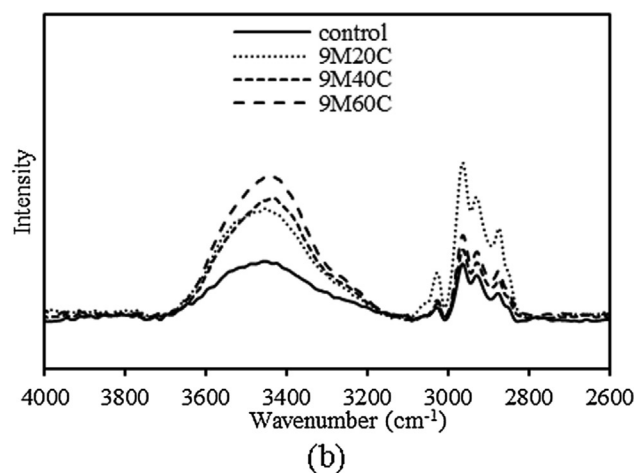
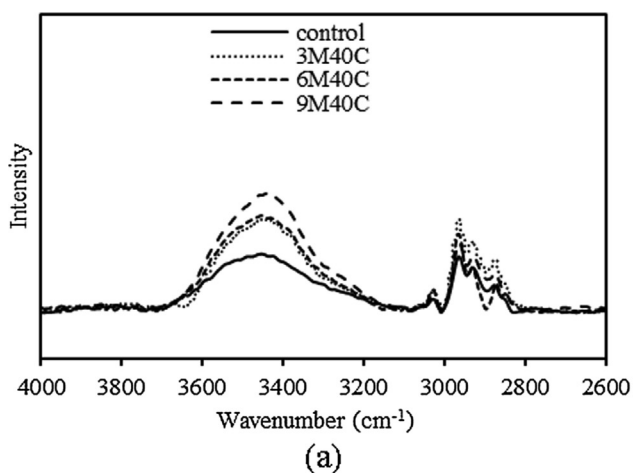


Fig. 18. FTIR spectra of BFRP bars conditioned in alkaline solution: (a) at 40 °C for different durations, (b) for 9 months at different temperatures.

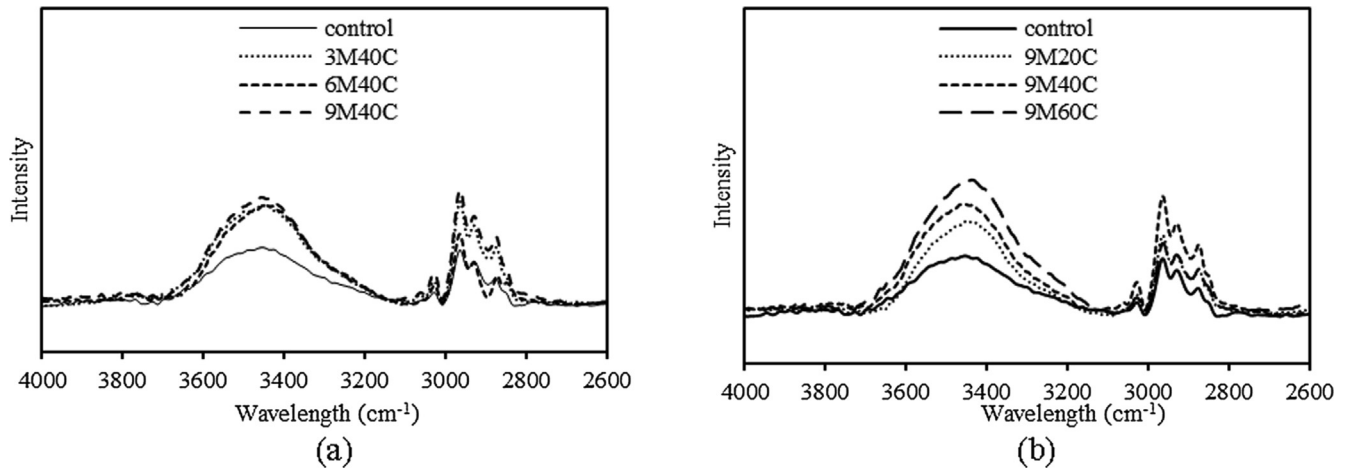


Fig. 19. FTIR spectra of BFRP bars embedded in concrete: (a) at 40 °C for different durations, (b) for 9 months at different temperatures.

two main spectral zones: 2800–3000 cm^{-1} , representing carbon-hydrogen (CH) groups, and 3200–3600 cm^{-1} , signifying hydroxyl (OH) groups. In the event of hydrolysis reaction, OH^- reacts with weak ester bonds in the polymer matrix, resulting in microcracks, accelerated moisture diffusion and debonding of the fibers [32,33]. As such, the infrared band of OH increases due to the formation of new hydroxyl groups, while that of CH remains unaffected. For this reason, the maximum peak for each infrared band was recorded. The ratio (OH/CH) is typically used to evaluate the relative amount of hydroxyl groups in the BFRP specimens [34–37].

Table 4 summarizes the OH-to-CH ratio of control and conditioned samples. The OH/CH of the control BFRP specimen was found to be 0.98. Up to 9 months of conditioning at 40 °C, no substantial change in the OH/CH was noted, indicating an insignificant increase in the amount of hydroxyl groups in the matrix and corroborating limited to no chemical degradation in BFRP bars when exposed to such conditioning schemes. At the elevated temperature of 60 °C, a slight increase of approximately 3.5% in the OH/CH was noted after 9-month of conditioning in the alkaline solution. For the concrete-encased BFRP specimens, the OH-to-CH ratio attained 0.995 after 9 months at 60 °C, representing a 1.12% increase over that of the unconditioned control specimen. Clearly, OH/CH did not significantly change when the BFRP specimens were conditioned in moist concrete, indicating limited chemical degradation of the matrix. These results are consistent with those reported by Benmokrane et al. [6].

4.6. DSC analysis

The glass transition temperature (T_g) of control and conditioned BFRP reinforcing bars are shown in Table 5. For each specimen, two scans were carried out. Samples exposed to alkaline solution for longer durations at the same temperature showed lower T_g values in the first scan. In fact, the first scan T_g decreased to 112 °C after 9 months of conditioning in the alkaline solution at 40 °C indicating plasticization of the matrix, which is known to occur in epoxy-based FRP reinforcing bars upon moisture uptake [6]. Conditioning at a higher temperature of 60 °C further reduced the T_g to 107 °C. This is possibly because of the increased moisture uptake, which modified the matrix polymeric structure. Further, a comparison between A6M40C and A9M20C showed that BFRP specimens conditioned for shorter durations and higher temperatures experienced a higher reduction in T_g than that of specimens conditioned for longer durations and lower temperatures. This provides evidence to the more detrimental effect of conditioning temperature on the performance and microstructure of BFRP reinforcing bars.

Specimens encased in moist concrete showed also a decreasing trend of T_g in the first scan. BFRP samples conditioned at 40 °C for 3 months reported no reduction in T_g . It was only until 6 months of conditioning that the T_g dropped to 115 °C, signifying that the BFRP matrix became more prone to degradation within the first 6 months. At an extended duration of 9 months, the T_g was also

Table 4

Band ratios of conditioned and control BFRP bars.

Sample Designation	Conditioning			Peaks		OH/CH	Increase (%)
	Medium	Duration (months)	Temperature (°C)	OH	CH		
Control	–	–	Ambient	98.1	99.7	0.984	–
A3M40C	Alkaline	3	40	97.9	99.1	0.988	0.41
A6M40C	Alkaline	6	40	97.9	99.4	0.985	0.10
A9M20C	Alkaline	9	20	97.8	99.3	0.985	0.10
A9M40C	Alkaline	9	40	97.7	99.3	0.984	0.00
A9M60C	Alkaline	9	60	99.0	97.2	1.018	3.46
C3M40C	Concrete	3	40	97.6	99.2	0.984	0.01
C6M40C	Concrete	6	40	97.5	99.0	0.985	0.10
C9M20C	Concrete	9	20	98.1	99.0	0.991	0.71
C9M40C	Concrete	9	40	97.5	98.4	0.991	0.71
C9M60C	Concrete	9	60	98.0	98.5	0.995	1.12

*Increase in OH/CH of conditioned with respect to that of the unconditioned control.

Table 5

Glass transition temperature of conditioned and control BFRP bars.

Sample Designation	Conditioning			T _g (°C)	
	Medium	Duration (months)	Temperature (°C)	1st scan	2nd scan
Control	–	–	Ambient	120	120
A3M40C	Alkaline	3	40	118	120
A6M40C	Alkaline	6	40	116	119
A9M20C	Alkaline	9	20	118	121
A9M40C	Alkaline	9	40	112	120
A9M60C	Alkaline	9	60	107	120
C3M40C	Concrete	3	40	120	121
C6M40C	Concrete	6	40	115	120
C9M20C	Concrete	9	20	116	119
C9M40C	Concrete	9	40	115	120
C9M60C	Concrete	9	60	112	119

reported as 115 °C. Apparently, the limited increase in moisture uptake between 6 and 9 months resulted in little to no change in the T_g in the first scan. Conditioning at a higher temperature of 60 °C resulted in a first scan T_g value of 112 °C. While an increase in the conditioning temperature from 20 to 40 °C led to a 1 °C decrease in T_g, a slightly greater reduction of 3 °C in T_g was noted when the temperature increased from 40 to 60 °C. This demonstrated that the degradation of the matrix was accelerated as the conditioning temperature increased, as also reported by De'Nève and Shanahan [23]. It should be noted that, even though the decrease in the T_g and the increase in the OH/CH ratio (*FTIR analysis*) for the BFRP bars of the current study were not excessive, relative to those reported previously for GFRP bars conditioned in moist concrete [29], an appreciable reduction in the tensile strength was recorded. This signifies the sensitivity of the tensile strength to any slightest chemical degradation, i.e. hydrolysis reaction and plasticization.

During the second scan, moisture absorbed by the BFRP specimens evaporated, thus reversing the plasticizing effect induced by moisture uptake. As such, the T_g values of conditioned samples recorded in the second scan were similar to those of the unconditioned control specimen, signifying the absence of irreversible chemical degradation. Therefore, it could be concluded that the degradation mechanisms of the BFRP reinforcing bars tested herein were damage to the fiber-matrix interface, plasticization, hydrolysis and loss of the polymer matrix during conditioning. No irreversible chemical degradation was detected.

5. Master curves for service life prediction of BFRP bars

Arrhenius model assumes that the rate of degradation is accelerated with an increase in the temperature (Eq. (3)). Accelerated aging test results of the current study confirmed the validity of this assumption. Eqs. (4) and (5) are other forms of Eq. (3).

$$k = Ae^{-E_a/RT} \quad (3)$$

where:

k = degradation rate (1/time)
 A = constant related to material and degradation process
 E_a = activation energy
 R = universal gas constant
 T = temperature in kelvin

$$\frac{1}{K} = \frac{1}{A} e^{E_a/RT} \quad (4)$$

$$\ln \frac{1}{k} = \frac{E_a}{RT} - \ln(A) \quad (5)$$

Eq. (2) shows that the degradation rate K can be transformed to the inverse of the time required for material property to reach a given value. Eq. (3) indicates that the logarithm of the time required for material property to reach a given value is a linear function of $1/T$ with slope of E_a/R .

The service life of BFRP bars can be extrapolated for any temperature using Arrhenius plots for actual year-round environmental conditioning in a specific city. Accelerated test results of the current study along with Arrhenius concept were employed to develop a master curve to predict the service life of BFRP bars at the average annual temperature of the city of Dubai (28 °C). Modeling procedures are summarized as follows:

1. Plot the tensile strength retention versus the time of accelerated aging.
2. Perform regression analysis to determine the best-fit relationship for each set of data at a specific temperature.
3. Plot ($\ln$) the time needed to reach a particular tensile strength retention versus the inverse of temperature (i.e. Arrhenius-type relationship).
4. Determine (E_a/R) from the Arrhenius-type relationships developed in step 3.
5. Calculate the time shift factor (TSF) relative to the reference temperature expected in natural weathering.
6. Use the TSF to estimate the time needed to reach a specific tensile strength retention at the reference temperature expected in natural weathering.
7. Develop a master curve for prediction of service life of BFRP bars by plotting the tensile strength retention values versus the corresponding anticipated time at the reference temperature expected in natural weathering.

Fig. 20 shows master curves developed based on the data of the current study for service life prediction of the BFRP bars at the

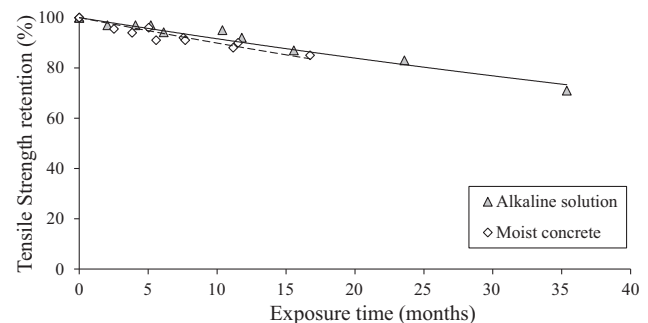


Fig. 20. Master curves for the BFRP bars at an annual average temperature of 28 °C.

average annual temperature of the city of Dubai (28 °C). These master curves are valid only for the tensile strength retention values and conditioning environments reported in this study. It can be seen that the master curves of both surrounding media (alkaline solution and moist concrete) are insignificantly different. This occurred because the tensile strength retention values reported in the current study for the BFRP specimens conditioned in both regimes at conditioning temperatures ≤ 40 °C were comparable. Fig. 20 also indicates that the tensile strength retention of the BFRP bars used in the current study would drop to 85% in approximately 1.5 years irrespective of the surrounding media. Based on the master curve of the bars conditioned in the alkaline solution, the tensile strength retention would drop to 71% in approximately 3 years.

The laboratory conditioning regimes adopted in the current study, which involved continuous immersion of concrete-encased BFRP specimens in water or direct exposure to an alkaline solution at high temperatures of up to 60 °C are considered very severe. In field conditions, BFRP bars in concrete are typically exposed to different levels of relative humidity not continuously immersed in water. As such, the developed master curves may be considered conservative. The second phase of this project would include testing of BFRP bar specimens embedded in concrete and exposed to field environmental conditions under different levels of sustained loads. Such conditioning regimes would better represent what might be encountered in practical settings.

6. Conclusions

This paper examined the durability performance BFRP bars exposed to alkaline solution and those embedded in moist concrete. The degradation mechanism induced by each conditioning regime is characterized by employing microstructure analysis techniques. Based on the experimental test results, the following conclusions are drawn:

1. The tensile strength retentions of the BFRP specimens conditioned in the alkaline solution at temperatures ≤ 40 °C were comparable to those of their counterparts conditioned in moist concrete irrespective of the time of exposure.
2. BFRP specimens conditioned in the alkaline solution at 60 °C for durations ≥ 6 months exhibited lower tensile strength retentions than those of their counterparts conditioned in moist concrete. After 9 months of conditioning at 60 °C, the tensile strength retentions of the BFRP specimens conditioned in the alkaline solution and moist concrete were 71 and 85%, respectively.
3. The impact of increasing the conditioning temperature on the tensile strength retention was affected by the conditioning duration and surrounding media. For the specimens conditioned in the alkaline solution, the reduction in the tensile strength caused by an increase in the conditioning temperature was more significant at the longer duration. The tensile strengths of the specimens conditioned in the alkaline solution at 60 °C for 3, 6, and 9 months were 4, 15, and 25% lower than those of their counterparts conditioned at 20 °C, respectively. In contrast, increasing the temperature from 20 to 60 °C for the specimens conditioned in moist concrete resulted in approximately 6.5% additional tensile strength reduction irrespective of the conditioning duration.
4. Moisture uptake was predominantly affected by the conditioning medium, duration, and temperature. BFRP specimens conditioned in the alkaline solution experienced generally higher moisture uptake than that of their counterparts encased in moist concrete. Increasing the conditioning duration and temperature led to an increase in moisture uptake, reduction in matrix retention, degradation of the fiber-matrix interfacial bond, and a decrease in tensile strength.
5. Conditioning induced changes to the microstructure of BFRP reinforcing bars. BFRP specimens experienced separation at the fiber-matrix interface, disintegration of the matrix, and formation of microgaps after conditioning durations ≥ 6 months in alkaline solution at temperatures of at least 40 °C. Circumferential fiber debonding and matrix disintegration were aggravated at the higher conditioning temperature of 60 °C. Concrete-encased BFRP specimens experienced less deterioration to the microstructure even at higher temperatures and longer durations. Apparently, conditioning BFRP bars in an alkaline solution for up to 6 months and at temperatures not exceeding 40 °C is capable of simulating conditioning in moist concrete under the same circumstances.
6. Chemical degradation in the matrix was detected by an increase in the amount of hydroxyl groups. Plasticization was noted as a reduction in the T_g corresponding to the first scan. Although conditioning-induced degradation to the microstructure was neither excessive nor easily detected, it affected the tensile strength. An irreversible chemical degradation was absent, evidenced by the second scan T_g . Specimens conditioned at higher temperatures and longer durations exhibited higher degree of degradation to the matrix with the former having a superior influence.
7. The accelerated aging test results along with Arrhenius concept were employed to develop master curves for service life prediction of the BFRP bars at the average annual temperature of the city of Dubai (28 °C). The master curves of both surrounding media (alkaline solution and moist concrete) were insignificantly different. In addition to being limited to the tensile strength retention values and conditioning environments reported in this study, the curves may be considered conservative because of the severity of the laboratory environmental conditioning regimes adopted in this research. The second phase of this project would include concrete-encased BFRP specimens exposed to field environmental conditions under a sustained load to better represent what might be encountered in practical settings.

Declaration of Competing Interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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References

- [1] ACI 440.1R, Guide for the Design and Construction of Concrete Reinforced with FRP bars, American Concrete Institute, Farmington Hills, Michigan, 2015.
- [2] ACI 440.3R, Guide Test Methods for Fiber-reinforced Polymers (FRPs) for Reinforcing or Strengthening Concrete Structures, American Concrete Institute, Farmington Hills, Michigan, 2012.
- [3] FIB, FRP Reinforcement in RC Structures, Bulletin 40, The International Federation for Structural Concrete FIB, Lausanne, Switzerland, 2007.
- [4] G. Wu, Z.-Q. Dong, X. Wang, Y. Zhu, Z.-S. Wu, Prediction of long-term performance and durability of BFRP bars under the combined effect of sustained load and corrosive solutions, *J. Compos. Constr.* 19 (3) (2015).
- [5] H.V.S. GangaRao, N. Taly, P.V. Vijay, Reinforced Concrete Design with FRP Composites, first ed., CRC Press, Boca Raton, FL, 2006.

- [6] B. Benmokrane, F. Elgabbas, E. Ahmed, P. Cousin, Characterization and comparative durability study of glass/vinylester, basalt/vinylester, and basalt/epoxy FRP bars, *J. Compos. Constr.* 19 (6) (2015) 04015008.
- [7] A.H. Ali, H.M. Mohamed, B. Benmokrane, A. ElSafty, O. Chaallal, Durability performance and long-term prediction models of sand-coated basalt FRP bars, *Compos. B: Eng.* 157 (2019) 248–258.
- [8] F. Elgabbas, E.A. Ahmed, B. Benmokrane, Physical and mechanical characteristics of new basalt-FRP bars for reinforcing concrete structures, *Constr. Build. Mater.* 95 (2015) 623–635.
- [9] G. Wu, X. Wang, Z. Wu, Z. Dong, Q. Xie, Degradation of basalt FRP bars in alkaline environment, *Sci. Eng. Compos. Mater.* 22 (6) (2015) 649–657.
- [10] A. Serbescu, M. Guadagnini, K. Pilakoutas, Mechanical characterization of basalt FRP rebars and long-term strength predictive model, *J. Compos. Constr.* 19 (2) (2015) 04014037.
- [11] F. Guo, S. Al-Saadi, R.K. Singh Raman, X.L. Zhao, Durability of fiber reinforced polymer (FRP) in simulated seawater sea sand concrete (SWSSC) environment, *Corros. Sci.* 141 (2018) 1–13.
- [12] X. Wang, L. Jiang, H. Shen, Z. Wu, Long-term performance of pultruded basalt fiber reinforced polymer profiles under acidic conditions, *J. Mater. Civ. Eng.* 30 (6) (2018) 04018096.
- [13] J. Shi, X. Wang, L. Ding, Z. Wu, Degradation of creep behaviors of basalt fiber-reinforced polymer tendons in salt solution, *J. Mater. Civ. Eng.* 30 (12) (2018) 04018317.
- [14] ASTM, Standard Test Methods for Constituent Content of Composite Materials, D3171, ASTM, West Conshohocken, PA, 2015.
- [15] ASTM, Standard Test Methods for Void Content of Reinforced Plastics, D2734, ASTM International, West Conshohocken, PA, 2016.
- [16] ASTM, Standard Test Method for Compressive Strength of Cylindrical Concrete Specimens, C39, ASTM, West Conshohocken, PA, 2015.
- [17] ASTM, Standard Test Method for Splitting Tensile Strength of Cylindrical Concrete Specimens, C496, ASTM, West Conshohocken, PA, 2011.
- [18] ASTM, Standard Test Method for Pulse Velocity Through Concrete, C597, ASTM, West Conshohocken, PA, 2016.
- [19] ASTM, Standard Test Method for pH of Aqueous Solutions With the Glass Electrode, E70, ASTM International, West Conshohocken, PA, 2015.
- [20] ASTM, Standard Test Method for Water Absorption of Plastics, D570, ASTM, West Conshohocken, PA, 2010.
- [21] H. El-Hassan, T. El Maaddawy, Microstructure characteristics of GFRP reinforcing bars in harsh environment, *Adv. Mater. Sci. Eng.* 2019 (2019) 19.
- [22] ASTM, Standard Test Method for Assignment of the Glass Transition Temperatures by Differential Scanning Calorimetry, E1356, ASTM, West Conshohocken, PA, 2008.
- [23] B. De Nève, M.E.R. Shanahan, Water absorption by an epoxy resin and its effect on the mechanical properties and infra-red spectra, *Polymer* 34 (24) (1993) 5099–5105.
- [24] T. D'Antino, M.A. Pisani, C. Poggi, Effect of the environment on the performance of GFRP reinforcing bars, *Compos. B: Eng.* 141 (2018) 123–136.
- [25] C. Scheffler, T. Förster, E. Mäder, G. Heinrich, S. Hempel, V. Mechtcherine, Aging of alkali-resistant glass and basalt fibers in alkaline solutions: evaluation of the failure stress by Weibull distribution function, *J. Non-Cryst. Solids* 355 (52) (2009) 2588–2595.
- [26] A.H.I. Mourad, B.M. Abdel-Magid, T. El-Maaddawy, M.E. Grami, Effect of seawater and warm environment on glass/epoxy and glass/polyurethane composites, *Appl. Compos. Mater.* 17 (5) (2010) 557–573.
- [27] A. Debaiky, G. Nkurunziza, B. Benmokrane, P. Cousin, Residual tensile properties of GFRP reinforcing bars after loading in severe environments, *J. Compos. Constr.* 10 (5) (2006) 370–380.
- [28] Y.A. Al-Salloum, S. El-Gamal, T.H. Almusallam, S.H. Alsayed, M. Aqel, Effect of harsh environmental conditions on the tensile properties of GFRP bars, *Compos. B Eng.* 45 (1) (2013) 835–844.
- [29] H. El-Hassan, T. El-Maaddawy, A. Al-Sallamin, A. Al-Saidy, Performance evaluation and microstructural characterization of GFRP bars in seawater-contaminated concrete, *Constr. Build. Mater.* 147 (2017) 66–78.
- [30] ASTM, Standard Test Methods for Loss-On-Drying by Thermogravimetry, E1868, ASTM, West Conshohocken, PA, 2015.
- [31] H. Kim, Y. Park, Y. You, C. Moon, Short-term durability test for GFRP rods under various environmental conditions, *Compos. Struct.* 83 (1) (2009) 37–47.
- [32] J. Chin, T. Nguyen, K. Aouadi, Effects of environmental exposure on fiber-reinforced plastic (FRP) materials used in construction, *Undefined* 19(4) (1997) 205–213.
- [33] G. Wu, X. Wang, Z. Wu, Z. Dong, Q. Xie, Degradation of basalt FRP bars in alkaline environment, *Sci. Eng. Compos. Mater.* (2015) 649.
- [34] M. Montaigu, M. Robert, E.A. Ahmed, B. Benmokrane, Laboratory characterization and evaluation of durability performance of new polyester and vinylester E-glass GFRP dowels for jointed concrete pavement, *J. Compos. Constr.* 17 (2) (2013) 176–187.
- [35] M. Robert, B. Benmokrane, Physical, mechanical, and durability characterization of preloaded GFRP reinforcing bars, *J. Compos. Constr.* 14 (4) (2010) 368–375.
- [36] M. Robert, B. Benmokrane, Effect of aging on bond of GFRP bars embedded in concrete, *Cem. Concr. Compos.* 32 (6) (2010) 461–467.
- [37] M. Robert, P. Cousin, B. Benmokrane, Durability of GFRP reinforcing bars embedded in moist concrete, *J. Compos. Constr.* 13 (2) (2009) 66–73.